

47. Determine whether the following matrices are diagonalizable over $\mathbb{R}$. If a matrix A is diagonalizable, provide an invertible matrix P and a diagonal matrix D such that $D = P^{-1}AP$. If a matrix A is non-diagonalizable, explain why.

(a) $A = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix}.$

(b) $A = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix}.$

(c) $A = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix}$

(These are the matrices from Problems 43 and 44. The starting point for the presentation should be a summary of eigenvalues / AMs and GMs / bases of eigenspaces for these matrices.)

48. For each of the following matrices, determine all values of the parameter a such that the matrix is diagonalizable.

(a) $A = \begin{pmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{pmatrix}.$

(b) $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ a & 2a & 3a \end{pmatrix}.$

49. Let $(a_0, a_1, a_2, \dots)$ be the sequence given by the following linear recurrence:

$$a_0 = 4, \quad a_1 = -1, \quad a_k = 3a_{k-1} + 10a_{k-2} \text{ for } k \geq 2.$$

Use matrix diagonalization to find an explicit formula for a_k .